

Adult Income Dataset

Project Overview

The Adult Census Income project is a binary classification task focused on predicting whether an individual's annual income exceeds \$50,000 with a dataset containing 48,000 records from U.S. Census data. The objective is to accurately categorize individuals into two classes: >50K or ≤50K. This analysis leverages a suite of demographic and employment features, including education, marital status, and occupation, to build a predictive model capable of identifying high-income earners (>50K) within the population.

Summary of the Approach

The approach for model training involves structured data preparation and modeling pipeline:

- **Data Cleaning:** The process involved identifying missing values (marked as "?" in the dataset) and replacing them with NaNs, followed by imputation using the mode for categorical data and the mean or median for numerical data.
- **Preprocessing:** Categorical variables (e.g., workclass, occupation) were scheduled for one-hot encoding, while binary variables (eg, gender, income) were targeted for label encoding. Numerical features like capital gain/loss and age were to be scaled using MinMaxScaler.
- **Validation Strategy:** The data was split into a 70% training set and a 30% testing set initially, and then the 30% was further divided into validation and testing ensuring the test data remained untouched until the final evaluation.
- **Model Selection:** The methodology involved training and comparing several classification models, specifically Logistic Regression, Decision Trees, and Random Forest.

Model Performance Analysis

The Gradient Boosting Classifier is the premier standalone model from the analysis. It has an Accuracy of 0.86 and a ROC-AUC of 0.92, it demonstrated the strongest individual capacity to distinguish between income classes, particularly benefiting from its iterative boosting approach which minimizes residual errors. However, overall winner among the models was The Bayesian Stacking Classifier. It is the optimal solution for this binary classification project. It achieved the highest recorded Accuracy of 0.8682, outperforming the standalone Gradient Boosting model. By aggregating predictions from Gradient Boosting, Random Forest, and Decision Trees, this

ensemble specifically reduces model variance and captures complex non-linear relationships better than any single algorithm, providing a more robust final vote when individual models express high uncertainty.

The Support Vector Classifier (SVC) was identified as the poorest performer. While its overall Accuracy was 0.8, its Recall was a critically low 0.28. It indicates a significant failure to handle the class imbalance inherent in the Adult Census dataset (where high-income earners represent the minority class). The SVC failed to learn the decision boundaries of the >50K class, resulting in an unacceptable volume of false negatives. This specific failure mode makes it ineffective for any application where identifying high-income earners is the primary objective.

Conclusion

To achieve high accuracy in predicting income levels, we used a three-step approach: cleaning the data to remove errors, standardizing the information, and combining several different AI models into one 'super-model.' This team-based approach allowed the strengths of each model to cancel out the others' weaknesses, resulting in a more reliable and accurate final prediction than any single method could provide on its own.